

1d_performance_predictors

Auto-generated from `2_pipeline/1d_performance_predictors/out/` on 2026-09-07 16:45.

Re-run `make_progress_md.py` after the script's output changes -- don't hand-edit this file.

Tables

Performance mi vs linear

protocol	predictor	n_trials	r	r2	p_linear	mi_bits_rav
DM	cur_stim_bin	3045	0.1515	0.02295	4.319e-17	0.01416
DM	prev_stim_bin	3027	-0.0345	0.00119	0.05774	0.0009082
DM	cur_runspeed	3045	0.1607	0.02582	4.633e-19	0.02754
DM	cur_pupil_area	988	0.05336	0.002848	0.09365	0.0235
DM	cur_motionenergy	2014	-0.0201	0.0004042	0.3672	0.01553
DM	cur_lick	3045	-0.06611	0.004371	0.0002616	0.009481
DM	prev_runspeed	3027	0.1291	0.01666	1.022e-12	0.01842
DM	prev_pupil_area	982	0.04882	0.002383	0.1263	0.02465
DM	prev_motionenergy	2003	-0.02849	0.0008115	0.2025	0.009303
DM	prev_lick	3027	-0.06437	0.004143	0.0003949	0.007184
DM	prev_correct	3027	0.2493	0.06214	4.202e-44	0.03057
DM	prev_lickResponse	3027	0.1419	0.02012	4.475e-15	0.01266
DP	cur_stim_bin	3194	0.1932	0.03732	3.155e-28	0.03667
DP	prev_stim_bin	3176	-0.009245	8.547e-05	0.6025	0.0001967
DP	cur_runspeed	3194	-0.02597	0.0006746	0.1422	0.009254
DP	cur_pupil_area	1137	-0.1712	0.0293	6.305e-09	0.03533
DP	cur_motionenergy	2085	-0.1116	0.01245	3.252e-07	0.03094
DP	cur_lick	3194	-0.06012	0.003614	0.0006752	0.002063
DP	prev_runspeed	3176	-0.04404	0.00194	0.01306	0.007671
DP	prev_pupil_area	1130	-0.1661	0.02758	1.957e-08	0.03569
DP	prev_motionenergy	2075	-0.1077	0.01161	8.716e-07	0.03284
DP	prev_lick	3176	-0.007104	5.046e-05	0.689	0.0003961
DP	prev_correct	3176	0.07635	0.00583	1.65e-05	0.004097

protocol	predictor	n_trials	r	r2	p_linear	mi_bits_rav
DP	prev_lickResponse	3176	0.1052	0.01106	2.814e-09	0.007849
DN	cur_stim_bin	6698	0.3275	0.1073	3.227e-167	0.1095
DN	prev_stim_bin	6672	0.01058	0.0001118	0.3878	0.003493
DN	cur_runspeed	6698	0.002196	4.821e-06	0.8574	0.00743
DN	cur_pupil_area	6212	-0.03416	0.001167	0.007084	0.009661
DN	cur_motionenergy	6431	-0.02451	0.0006008	0.04936	0.006698
DN	cur_lick	6698	-0.01938	0.0003756	0.1127	0.0004039

(showing first 30 of 36 rows -- see the linked CSV for the rest)

[performance_mi_vs_linear.csv](#)

Performance multivariate mi vs linear

protocol	predictors	n_predictors	n_trials	r2	adj_r2	p_linear	mi
DM	cur_stim_bin, prev_stim_bin, cur_runspeed, cur_pupil_area, cur_motionenergy, cur_lick, prev_runspeed, prev_pupil_area, prev_motionenergy, prev_lick, prev_correct, prev_lickResponse	12	979	0.1683	0.158	7.089e-32	0.0
DP	cur_stim_bin, prev_stim_bin, cur_runspeed, cur_pupil_area, cur_motionenergy, cur_lick, prev_runspeed, prev_pupil_area, prev_motionenergy, prev_lick, prev_correct, prev_lickResponse	12	1129	0.1016	0.09199	5.741e-20	0.1

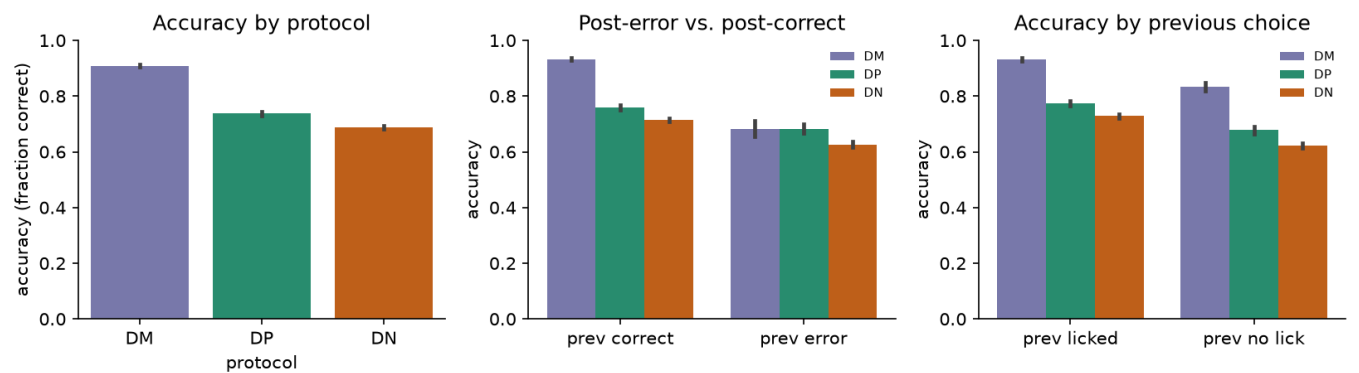
protocol	predictors	n_predictors	n_trials	r2	adj_r2	p_linear	mi
DN	cur_stim_bin, prev_stim_bin, cur_runspeed, cur_pupil_area, cur_motionenergy, cur_lick, prev_runspeed, prev_pupil_area, prev_motionenergy, prev_lick, prev_correct, prev_lickResponse	12	6188	0.1293	0.1276	2.085e-175	0.1

[performance_multivariate_mi_vs_linear.csv](#)

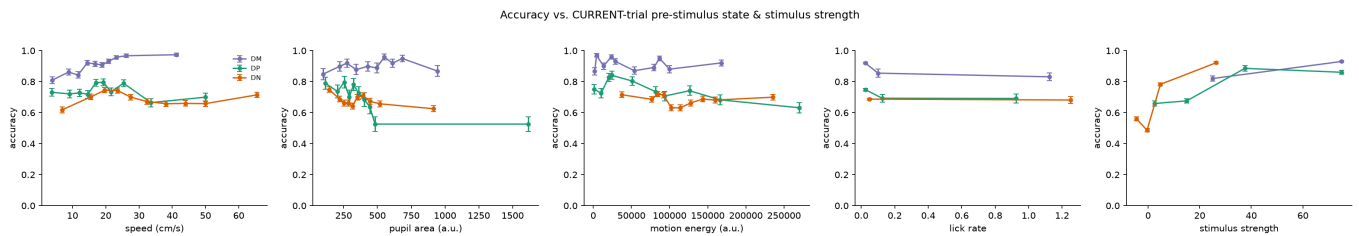
Figures

Performance overview

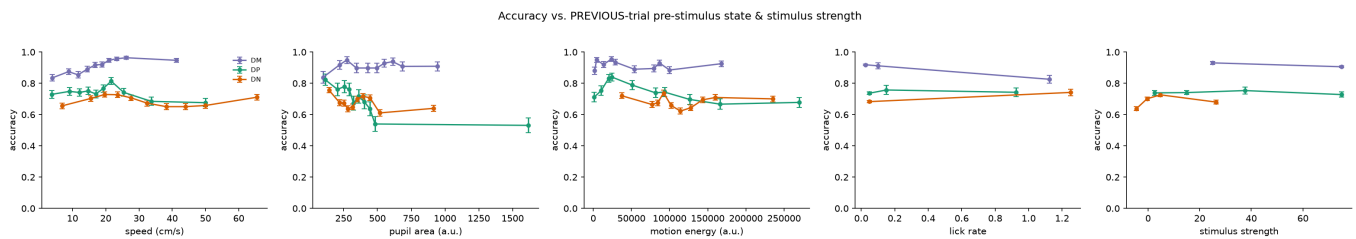
Performance overview (all trials, correctness generalized to every signal level)



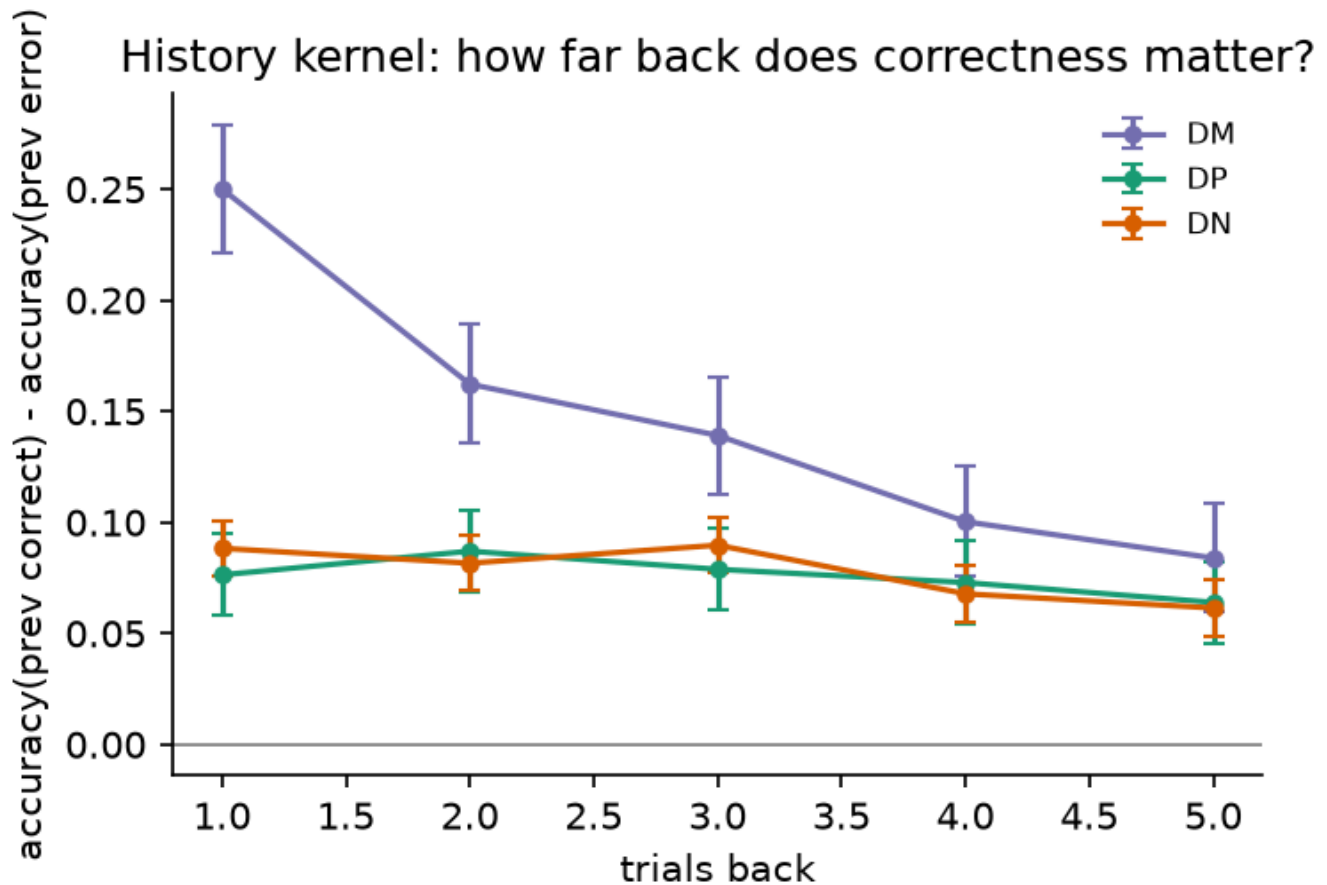
Tuning current trial



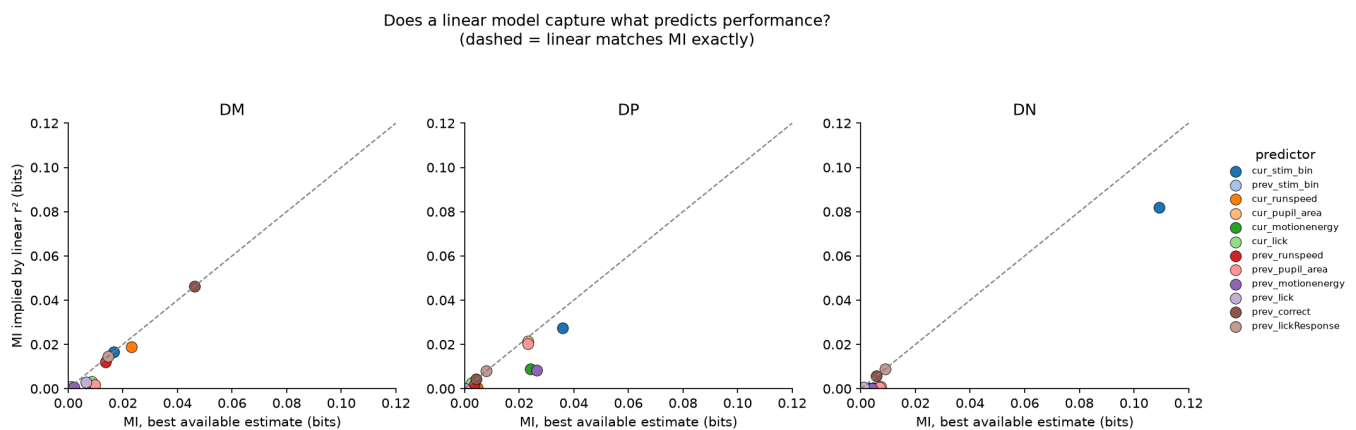
Tuning previous trial



History kernel

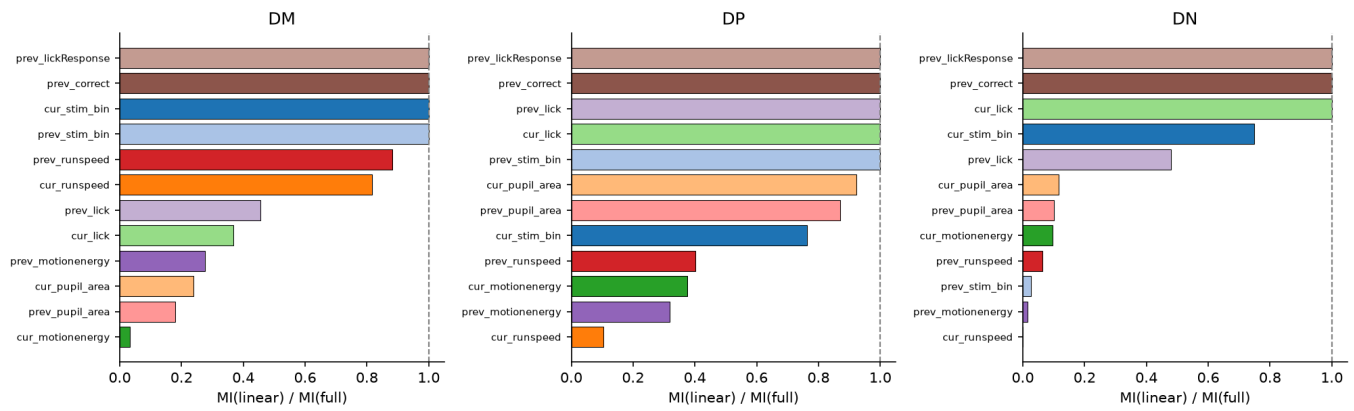


Mi vs linear summary



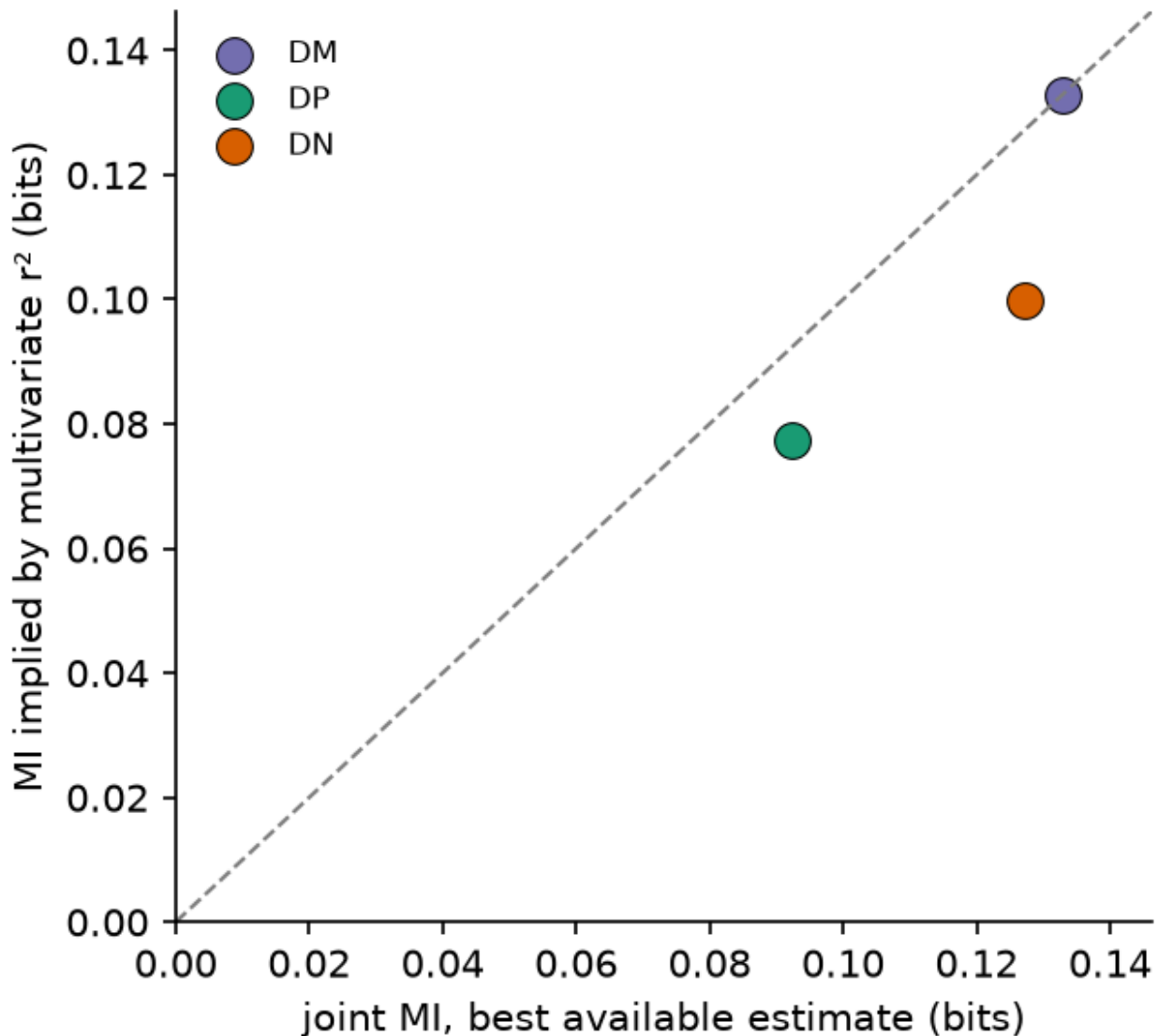
5a frac explained linearly

Fraction of mutual information a linear fit actually captures, per predictor



Multivariate mi vs linear summary

Multivariate: all predictors jointly vs. performance
(dashed = linear matches MI exactly)



6a multivariate frac explained linearly

Fraction of the joint MI a
multivariate linear fit captures

